

REVIEW & INTERPRETATION

A Simple Protocol for AMMI Analysis of Yield Trials

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ABSTRACT

The Additive Main effects and Multiplicative Interaction (AMMI) model has been used extensively for analysis of multi-environment yield trials for two main purposes: understanding complex genotype $\times$ environment interactions and increasing accuracy. Nevertheless, AMMI analyses implementing best practices have been rare. Accordingly, this paper presents a simple protocol with four steps: (i) analysis of variance, (ii) model diagnosis, (iii) mega-environment delineation, and (iv) agricultural recommendations. This protocol is illustrated with an international bread wheat trial. This paper concerns a basic and common application of AMMI: yield-trial analysis without consideration of special structure or additional data for either genotypes or environments. Best practices involve using both treatment and experimental designs to gain accuracy and exploiting both broad and narrow adaptations to increase yields.

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Abbreviations: AMMI, Additive Main effects and Multiplicative Interaction; AOV, analysis of variance; CIMMYT, International Maize and Wheat Improvement Center; df, degrees of freedom; E, environment main effects; G, genotype main effects; GE, genotype $\times$ environment interaction effects; GE_N , GE noise; GE_p , practical portion of GE_S ; GE_S , GE signal; IPC, interaction principal component; PCA, principal components analysis; RCB, randomized complete block; SS, sum of squares.

APPLICATIONS of the AMMI model to yield trials have appeared frequently during the last two decades, and there have been several recent review articles (Gauch, 2006; Yan et al., 2007; Gauch et al., 2008; Yang et al., 2009). Nevertheless, AMMI analyses implementing best practices have been rare. The unfortunate consequence is that valuable information in costly data is wasted, thereby diminishing agricultural insights and reducing yield gains. A simple protocol for using AMMI effectively is presented here with four steps: (i) analysis of variance, (ii) model diagnosis, (iii) mega-environment delineation, and (iv) agricultural recommendations. This protocol is illustrated with an international bread wheat (*Triticum aestivum* L.) yield trial (Crossa et al., 1991).

PRELIMINARY CONSIDERATIONS

A planned experiment typically has two designs, the treatment design and the experimental design. For a yield trial with a number of genotypes tested in a number of environments, its treatment design is a two-way factorial of genotypes by environments. Yield trials also often have an experimental design involving randomization, replication, blocking, or use of spatial information,

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with the randomized complete block (RCB) design being especially common.

The two main purposes of AMMI analysis of a yield trial's treatment design are: (i) understanding complex genotype $\times$ environment interactions, including delineating mega-environments and selecting genotypes to exploit narrow adaptations, and (ii) increasing accuracy to improve recommendations, repeatability, selections, and genetic gains. The main purposes of an experimental design are assigning experimental units to treatments, quantifying errors, and gaining accuracy.

Analysis of variance (AOV) of a yield trial's treatment design partitions its variance into three sources: genotype main effects (G), environment main effects (E), and genotype $\times$ environment interaction effects (GE). For agronomists manipulating environments, E and GE are relevant. For the present context of breeders manipulating genotypes, G and GE are relevant because only they affect genotype rankings. Traits commonly measured include yield and yield components, biotic and abiotic resistances, and nutritional quality. But for simplicity, the measured trait is termed "yield" here, and the experiment a "yield trial."

AMMI first applies AOV to partition the variation into G, E, and GE, and then it applies principal components analysis (PCA) to GE. Accordingly, both G and GE are analyzed, but separately and without confounding, which is crucial because they present plant breeders with different challenges and opportunities (Gauch et al., 2008). Although yield trials vary, commonly both G and GE are statistically significant and agriculturally important. Broad adaptations are associated with G and are beneficial everywhere, whereas narrow adaptations are associated with GE and require subdividing the environments into two or more mega-environments. A mega-environment is defined as a subset of the environments having the same, or at least similar, winning genotypes.

Crossover genotype $\times$ environment interactions involve changes in genotype rankings across environments. Crossovers between inferior genotypes are rather inconsequential. But crossovers between winners are of agricultural significance because they implicate narrow adaptations that prompt different genotype selections and recommendations across different mega-environments.

Applications of AMMI are diverse. First, the environments may have a special structure, such as a complete factorial of a number of locations tested over a number of years. Second, the genotypes may have a special structure, such as a diallel design. Third, a yield experiment may have additional data and purposes, such as environmental data for explaining crop responses, or genetic marker data for detecting relevant genes. Although many variations are possible, this brief paper concerns only one particularly simple and common application: yield-trial analysis without consideration of special structure or additional data

(which may nevertheless be present). This basic AMMI analysis is important, both because it is so common, and because researchers must master the basic case as a foundation for any of the more complicated applications. For instance, a preliminary AMMI analysis to gain accuracy can greatly improve subsequent analyses for generating hierarchical classifications (Smith and Gauch, 1992) or detecting quantitative trait loci (Gauch et al., 2011).

FIRST STEP: ANALYSIS OF VARIANCE

Three numbers from the AOV provide a preliminary indication whether AMMI analysis will be worthwhile: the sum of squares (SS) for genotypes (G), GE signal (GE_S), and GE noise (GE_N). The SS values for G and GE are direct outputs from AOV. To estimate the SS for GE_N , simply multiply the error mean square (from replication) by the number of degrees of freedom (df) for GE (Gauch, 1992, p. 147). Then obtain GE_S by subtracting GE_N from GE.

AMMI analysis is appropriate for datasets having substantial G and substantial GE_S . Especially when the SS for GE_S is at least as large as that for G, as happens frequently, AMMI analysis will probably be worthwhile. On the other hand, occasionally GE is buried in noise, with the SS for GE_N approximately equal to that for GE. In that case, GE should be ignored, so AMMI analysis is inappropriate.

SECOND STEP: MODEL DIAGNOSIS

AMMI constitutes a model family, not a single model. Consequently, model diagnosis is required to determine which member of this model family is best for a given dataset and research purpose. The AMMI model equation is:

$$Y_{ge} = \mu + \alpha_g + \beta_e + \sum_n \lambda_n \gamma_{gn} \delta_{en} + \rho_{ge} \quad [1]$$

where Y_{ge} is the yield of genotype g in environment e , μ is the grand mean, α_g is the genotype deviation from the grand mean, β_e is the environment deviation, λ_n is the singular value for interaction principal component (IPC) n and correspondingly λ_n^2 is its eigenvalue, γ_{gn} is the eigenvector value for genotype g and component n , δ_{en} is the eigenvector value for environment e and component n , with both eigenvectors scaled as unit vectors, and ρ_{ge} is the residual. Commonly the interaction scores are scaled as $\lambda_n^{0.5} \gamma_{gn}$ and $\lambda_n^{0.5} \delta_{en}$ so that their products estimate interactions directly, without the need of yet another multiplication by λ (Gauch, 1992, p. 85).

Successive IPCs are denoted by IPC1, IPC2, IPC3, and so on, and the number of these components is 1 less than the minimum of the number of genotypes and number of environments. The member of the AMMI model family retaining 0 components is denoted by AMMI0, and the following members retaining 1 or more components are denoted by AMMI1, AMMI2, AMMI3, and so

on, up to the full model retaining all components denoted by AMMIF. The fitted values of the full model automatically equal the raw data Y_{ge} exactly, so the residual term disappears. But reduced models leave a residual ρ_{ge} .

A yield trial with an experimental design has additional terms in its model equation beyond those for only the treatment design. For instance, the equation for the AMMI model applied to a yield trial with the popular RCB experimental design is:

$$Y_{ger} = \mu + \alpha_g + \beta_e + \sum_n \lambda_n \gamma_{gn} \delta_{en} + \rho_{ge} + \kappa_{r(e)} + \varepsilon_{ger} \quad [2]$$

where Y_{ger} is the yield of genotype g in environment e for replicate r , and the two additional terms beyond those in Eq. [1] are $\kappa_{r(e)}$ which is the block effect for replication r within environment e , and ε_{ger} which is the error. For the RCB design, the yields Y_{ge} of the raw data AMMIF are simply the averages over the R replicates, $(\sum_r Y_{ger})/R$, although some other experimental designs make adjustments to the raw data.

Three criteria for model diagnosis are found in the AMMI literature: statistical significance, agricultural interpretability, and predictive accuracy. It is possible to have different criteria for different agricultural purposes, even for a given dataset, and hence to use more than one member of the AMMI model family, including the raw data, AMMIF.

Statistical Significance

F-tests using Gollob's df assignments have been used extensively because this test is easy to apply (Gollob, 1968), but these tests are well known to diagnose too many components. More complicated F-tests have been developed that better optimize predictive accuracy, especially F_R -tests (Piepho, 1995). A fundamental problem with significance tests is that their sensitivity depends on the size of the dataset (Raftery, 1995). Consequently, statistical significance may have little bearing on other matters of great concern, such as predictive accuracy and agricultural interpretability, unless the statistic was intentionally developed as a surrogate for predictive accuracy.

Agricultural Interpretability

Components having an evident agricultural interpretation, such as IPC1 associating with rainfall, are more useful for some purposes than those that are merely statistically significant. When a given IPC has a large SS, and especially when this SS equals or surpasses the SS for G , the underlying causal factor is usually already well known to agricultural researchers (Gauch, 1992, p. 231–236). Also, correlation coefficients can be calculated between IPC scores and various genetic or environmental measurements to clarify causal factors. Nevertheless, agricultural or ecological interpretations of patterns in AMMI parameters are sometimes rather subjective.

Predictive Accuracy

Of these three criteria for model diagnosis, predictive accuracy merits special attention. Predictive accuracy has mostly been assessed within yield trials by data splitting and cross validation, provided that replicated data are available (Gauch, 1992, p. 134–153). Fortunately, such within-trial accuracy has been demonstrated by extensive experiments to contribute to between-trial accuracy for new years and new locations (Annicchiarico et al., 2006; Ebdon and Gauch, 2011). The usual cross validation method in AMMI analysis involves selecting one (or more) replicate at random for each individual genotype-environment combination to serve as validation data, and the remaining data are used to fit the AMMI model family. Then the root mean square prediction difference between validation data and model estimates is calculated for each member of the AMMI model family (or a subset of interest, such as AMMI0 through AMMI5 and also AMMIF), with small values indicating predictive accuracy. Often this procedure is repeated (say, 1000 times) with different randomizations and the results averaged for more reliable results. Cross validation is included in the MATMODEL software for AMMI analysis, which is available for free (Gauch, 2007). Incidentally, there are several other cross-validation schemes, such as the leave-one-out method, that can be applied to data matrices without having or considering replication (Bro et al., 2008).

The data are a mixture of signal, due to treatment effects and agriculturally important causal factors, and noise, due to measurement errors and uncontrolled factors. Ordinarily the signal is relatively simple compared with the complex and idiosyncratic noise. Then early IPCs selectively recover signal, whereas late components selectively recover noise. Recovery of signal improves predictions, whereas recovery of noise degrades predictions. When the predictive accuracy of a sequence of increasingly complex (decreasingly parsimonious) AMMI models is graphed from left to right, accuracy initially increases while mostly signal is captured, and thereafter decreases while mostly noise is captured, resulting in a response called Ockham's hill (MacKay, 1992; Gauch, 1988, 1992, p. 134–153, 2006b, 2012, p. 174–198). To the left of the peak on Ockham's hill, models are less accurate because they underfit signal; to the right, they are less accurate because they overfit noise. It is better to have a modest amount of both problems than a huge amount of either problem.

By selecting the AMMI model at the peak of Ockham's hill, AMMI provides an opportunity to gain accuracy. AMMI gains accuracy by partitioning the treatment design into a signal-rich model and a discarded noise-rich residual. Often the accuracy gain from a parsimonious AMMI model is equivalent to increasing the number of replications by a factor of 2 to 4, which is larger than typical accuracy gains from statistical analysis of blocks or spatial information.

Cross validation can characterize Ockham's hill in its entirety, thereby clarifying the tradeoff between parsimony and accuracy for a given dataset. If the peak of Ockham's hill is fairly broad, as occurs frequently, then a small sacrifice in accuracy can enable a considerable gain in parsimony. In the moderately common case that two successive members of the model family are nearly tied for maximizing predictive accuracy, say AMMI1 and AMMI2, in general it is preferable to select the simpler model, especially if IPC1 is agriculturally interpretable but IPC2 is not.

An alternative to cross validation, which is computationally trivial and yet rather reliable, is to estimate the SS for GE_s as explained above and then retain as many IPCs as needed for the sum of their eigenvalues to approximately equal the SS for GE_s (while preferring the more parsimonious lower-order model if this criterion lands nearly midway between two models). When both cross validation and this remarkably simple diagnostic are applied to a given dataset, the diagnoses are often identical, or at least close (which is quite adequate in the common case that Ockham's hill has a broad peak).

Regrettably, the agricultural literature using AMMI (and related statistical methods) has been dominated by papers that fail to implement any model diagnosis whatsoever, and instead take AMMI1 (or its closest equivalent in related methods) as an automatic default because convenient two-dimensional graphs can be based on such models. However, the contemporary AMMI literature is rightly insistent on the necessity of model diagnosis to implement best practices (Gauch et al., 2008; Yang et al., 2009).

Besides gaining accuracy from AMMI analysis of the treatment design, the more familiar means for agricultural researchers to increase accuracy involves the experimental design, using RCB or other designs. Because the treatment and experimental designs have different df, both provide opportunities to gain accuracy. Statistical analysis of a RCB experimental design makes adjustments for individual replicates Y_{ger} and ordinarily reduces the error mean square. However, a RCB design does not make any adjustments to the averages over replicates Y_{ge} and hence neither adjusts yield estimates closer to their true values nor changes genotype rankings. By contrast, AMMI analysis of a yield trial's treatment design can adjust yield estimates closer to their true values on average (as proven empirically by cross validation), which changes and improves genotype rankings. But some other experimental designs, including ones using incomplete blocks, do make beneficial adjustments in Y_{ge} estimates.

THIRD STEP: MEGA-ENVIRONMENT DELINEATION

When there are crossovers between winning genotypes, subdivision of a growing region into two or more mega-environments is necessary for exploiting the narrow adaptations that often provide a substantial opportunity to

increase yield, sometimes even more so than the broad adaptations. There are three important considerations.

First, model diagnosis is essential because as the selected member of the AMMI model family changes, the mega-environments also change, with higher-order models tending to define a larger number of mega-environments. For instance, Gauch et al. (2008) presents an example for which AMMI1 delineates 2 mega-environments, AMMI2 delineates 3, AMMI3 delineates 4, and AMMI4 to the full model AMMI8 delineate 5 or 6. Consequently, mega-environments cannot be delineated meaningfully or reliably without first performing a model diagnosis to select the best member of the AMMI model family for a given dataset. Model diagnosis enables researchers to distinguish between GE_s causing actual narrow adaptations and GE_N generating spurious complexity.

Second, it is important for mega-environments to have predictive potential for locations and years beyond those in the dataset at hand (Piepho and Möhring, 2005). This predictive role is greatly enhanced if the mega-environments, initially delineated in terms of winning genotypes, have an evident environmental interpretation in terms of environmental gradients or geographical subregions. Predictable environmental factors associated with locations or management practices increase the number of usable mega-environments, whereas unpredictable environmental factors associated with years decrease the number and usefulness of mega-environments (Gauch, 1992, p. 217–230; Annicchiarico, 1997, 2002; Annicchiarico et al., 2006).

Third and finally, because subdivision into several mega-environments is costly for breeding programs and seed distributors alike, only a practical portion GE_p of the interaction signal GE_s may be available for exploiting narrow adaptations to increase yields. This limitation necessitates selecting a low-order model such as AMMI1 for delineating a small and manageable number of mega-environments, even though a higher model such as AMMI3 may be selected for the different research purpose of optimizing predictive accuracy. But fortunately, merely 2 or 3 mega-environments often suffice to allow GE_p to capture a sizable portion of GE_s . When GE_p is sizable relative to G , the benefits from exploiting narrow adaptations are likely to outweigh the costs from subdivision of a growing region or breeding program into a few mega-environments.

Mega-environments can be displayed by both tables and graphs. A ranking table shows the ranks for the best several genotypes in each environment. Listing the environments in order by their IPC1 scores (but not by their means) makes these tables more structured and informative, especially when IPC1 has an evident agricultural interpretation, such as a gradient from short to long growing seasons. Tables have three advantages. First, ranking tables can show any member of the AMMI model family equally readily, whereas graphs can accommodate only

AMMI1 and AMMI2 (Gauch and Zobel, 1997; Gauch et al., 2008). Second, ranking tables can identify as many of the best genotypes as desired, whereas mega-environment graphs identify only the top ranked genotype for each environment. Third, a single ranking table can list several AMMI models side by side to facilitate comparisons and to serve multiple research purposes. This is useful in the frequent case that several members of the AMMI model family are all of interest for a given dataset, such as AMMI1 for mega-environment delineation, AMMI3 for accuracy gain, and AMMI5 for representing the raw data (as in the following example of a wheat yield trial). Nevertheless, biplot and mega-environment graphs using AMMI1 and AMMI2 can also be helpful for visualizing complex patterns in yield-trial data, so graphs and tables are complementary, having somewhat different strengths.

FOURTH STEP: SELECTION AND RECOMMENDATION

A major purpose of yield-trial research is selection of the best genotypes to retain in a breeding program or recommendation of the best cultivars for a growing region. But selection tasks are remarkably difficult because noise obscures true winners, generates spurious complexity, and reduces genetic gains (Gauch, 1992, p. 171–204).

A common approach for genotype selection pursues both high yield and stability. But this approach has five considerable problems. First, there are dozens of stability parameters, making any particular choice difficult to justify, even though they can be organized into four groups (Annicchiarico, 2002, p. 79–87). However, a specific stability concept of interest is stability across years within a given location or mega-environment because it reduces susceptibility to unpredictable genotype $\times$ environment interactions, and yet it also acknowledges opportunity from predictable interactions. Second, there are manifold ways to integrate high yield and stability, but many fail to optimize any known and agriculturally significant outcome. Third, stability is a meaningful objective only within an individual mega-environment, not across multiple mega-environments, and yet much of the literature ignores this crucial requirement. Indeed, selecting for stability across mega-environments amounts to sacrificing potential yield gains from narrow adaptations. Fourth, at least 8 trials within each mega-environment are necessary for reasonably reliable estimates of stability (Annicchiarico, 2002, p. 82). Yet researchers often use fewer trials. Fifth and finally, regarding instability solely as a problem to be minimized is a defective paradigm because the complete story is that instability (GE) presents plant breeders with both problems and opportunities. Remarkably little of the literature on stability has adequately addressed all five of these concerns.

Given these five problems, other approaches for selection and recommendation merit consideration. A popular

alternative with tremendous advantages is to determine which genotypes win in which environments according to a parsimonious AMMI model that is more accurate than the raw data. Because AMMI analysis excludes much of the noise in the raw data that generates spurious complexity devoid of repeatability, it results in a smaller and more manageable roster of winning genotypes. What-wins-where is a straightforward concept of direct, immediate agricultural concern (Gauch, 1992, p. 205–240; Gauch and Zobel, 1997; Gauch et al., 2008).

EXAMPLE

The protocol for basic AMMI analysis outlined above can be illustrated with an international bread wheat trial for grain yield (t ha^{-1}) having 18 genotypes grown in 25 environments (Crossa et al., 1991). This experiment had a RCB design with 3 replications, but only the averages over replications are available from that publication, as well as the error mean square. The International Maize and Wheat Improvement Center (CIMMYT) conducted this trial in Algeria, Bangladesh, Cyprus, Egypt, India, Jordan, Kenya, Mexico, Nepal, Pakistan, Saudi Arabia, Sudan, Spain, Tunisia, and Thailand. The 25 environments were clustered into merely 2 mega-environments. Environment IPC1 scores from the AMMI1 analysis were associated with a gradient from short to long growing seasons, and correspondingly genotype IPC1 scores were associated with a trend from early to late maturing genotypes. For details on these genotypes and environments, consult Crossa et al. (1991). This brief example is not intended as a comprehensive recapitulation or illustration of every kind of statistic and graph and procedure discussed above, but rather is intended to show the main considerations required for effective AMMI analysis. In particular, an AMMI1 biplot is not shown here because it has already been published as Fig. 3 in Crossa et al. (1991).

First Step

Table 1 presents the AOV for AMMI5. The sources listed in this table correspond to the terms in Eq. [2] for the AMMI model applied to a yield trial with a RCB experimental design. The df for IPCs were assigned according to the method of Gollob (1968). The SS for G is 54.337, for GE_N is $408 \times 0.438 = 178.704$, and for GE_S is $397.527 - 178.704 = 218.823$ (and for the irrelevant E is 7108.372). Since GE_S is 4 times as large as G, AMMI analysis is quite likely to be worthwhile. For the present context of plant breeding, 96.4% of the treatment SS is irrelevant, namely E and GE_N , whereas 3.6% is relevant, namely G and GE_S .

Incidentally, another popular analysis of multi-environment yield trials has been linear regression of genotypes on environment means (Finlay and Wilkinson, 1963), which preceded AMMI analysis by about two decades (Kempton, 1984). However, whereas IPC1 captures a SS

Table 1. Analysis of variance for the AMMI5 model of grain yield (t ha⁻¹) for an international bread wheat trial. The yield data used for this analysis are from Table 6 in Crossa et al. (1991), and the values for blocks and error are from their Table 5. Indenting of the sources indicates subtotals.[†]

Source	df	SS	MS
Total	1349	8028.236	5.951
Treatment Design	449	7560.236	16.838
Genotype	17	54.337	3.196
Environment	24	7108.372	296.182
GE Interaction	408	397.527	0.974
IPC1	40	105.963	2.649
IPC2	38	54.849	1.443
IPC3	36	46.679	1.297
IPC4	34	42.068	1.237
IPC5	32	33.572	1.049
Residual	228	114.396	0.502
Experimental Design	900	468.0	0.520
Blocks within Env.	50	96.0	1.920
Error	850	372.0	0.438

[†]df, degrees of freedom; SS, sum of squares; MS, mean square.

of 105.963, which is 26.7% of GE or 48.4% of GE_S, these regressions capture a SS of only 37.815, so they are rather ineffective for this complex dataset. Adaptive responses of the genotypes can be depicted by linear regression lines in a graph with the environment means ($\mu + \beta_e$) on the abscissa and nominal yields (which omit environment deviations β_e) on the ordinate, as in Fig. 2 in Finlay and Wilkinson (1963). However, AMMI1 provides an analogous graph of adaptive responses that routinely captures much more GE by substituting environment IPC1 scores ($\lambda_1^{0.5} \delta_{e1}$) on the abscissa, as in Fig. 3 in Gauch et al. (2008). Also see Fig. 7 and 11 in Gauch et al. (2011).

The results in Table 1 illuminate the relative importance of the two opportunities that plant breeders have for increasing accuracy: (i) the treatment design, and (ii) the experimental design. As explained in the next step, model diagnosis for this wheat dataset selects AMMI3 for gaining accuracy. AMMI3 analysis applied to the treatment design discards most of the deleterious GE_N with its SS of 178.704, and statistical analysis of the RCB experimental design discards the deleterious block effects with their considerably smaller SS of 96.0. Besides this quantitative difference, there is also the qualitative difference that statistical analysis of the treatment design by a suitable AMMI model does adjust yield estimates closer to their true values on average, whereas statistical analysis of the RCB experimental design does not. The mean square residual of AMMI3 is 0.646, and hence its root mean square adjustment to AMMIF yield estimates based on a mean of 3 replicates is $(0.646/3)^{0.5} = 0.464$, or 10% of the grand mean of 4.65378 t ha⁻¹, which causes numerous changes in genotype rankings. AMMI1 makes larger adjustments of 11%. In perspective, adjustments

on the order of 10% to yield estimates based on the raw data AMMIF are clearly warranted because the standard error of these means over 3 replicates is $(0.438/3)^{0.5} = 0.382$ or 8.2% of the grand mean (or 8.9% if the RCB design is ignored). Regrettably, the dominant practice in contemporary yield-trial research is to exploit only the experimental design for gaining accuracy. This prevalent departure from best practices exacts a substantial opportunity cost in terms of slower progress from plant breeding (Gauch, 1992, p. 178–185, 2012, p. 13–14, 187–189).

Second Step

Model diagnosis for AMMI is routinely done using cross validation for experiments with replication, but this is not an option for this wheat trial because the replicated data were not published. But the error mean square was published, which can guide model diagnosis by retaining as many IPCs as needed for the sum of their eigenvalues to approximate the SS for GE_S, namely 218.823. From Table 1, the AMMI3 model captures 207.491, which is close (and it is better not to add IPC4 with its SS of 42.068). Hence, the model diagnosis for optimizing predictive accuracy is AMMI3.

Incidentally, Table 3 in Crossa et al. (1991) uses results from cross validation of the replicated data to diagnose AMMI1 as being most predictively accurate for this dataset, which is contrary to the present diagnosis of AMMI3. However, subsequently Piepho (1994) showed that cross validation should be modified to account for a RCB design, thereby rendering that previous diagnosis questionable since it is prone to selecting an excessively low-order model. The present diagnosis of AMMI3 is more reliable because the RCB experimental design has been accounted for in Table 1 by removing the variation for blocks within environments from the estimate of the error mean square. Although AMMI3 achieves somewhat greater accuracy than AMMI1, it remains that AMMI1 is still much more accurate than the raw data AMMIF and therefore AMMI1 greatly clarified the results in Crossa et al. (1991).

Third Step

Mega-environment delineation is the next step. Table 2 is a ranking table listing the top 5 genotypes in each of the 25 environments according to 3 members of the AMMI model family: AMMI1, AMMI3, and AMMIF. These models are of particular interest because AMMI1 is suitable for mega-environment delineation, AMMI3 optimizes predictive accuracy, and AMMIF represents the raw data. According to AMMIF, the raw data, 12 of the 18 genotypes win first rank in at least 1 environment. And according to AMMI3, the most accurate member of this family, there are 8 winners, which is still unwieldy. But AMMI1 has only 3 winners, Genotypes 5, 9, and 15. Furthermore, by accepting merely 3 second-place winners in the AMMI1 rankings (in environments 3, 17, and 20), the genotype

Table 2. A ranking table showing the top 5 genotypes according to 3 members of the Additive Main effects and Multiplicative Interaction (AMMI) model family. AMMI1 facilitates mega-environment delineation, AMMI3 optimizes predictive accuracy, and AMMIF represents the raw data. The 25 environments are ordered by their IPC1 scores, which reflect an ecological gradient from short growing seasons at the top to long growing seasons at the bottom, and the blank line separates the 2 mega-environments delineated by AMMI1S. Winners of particular interest, early maturing Genotype 5 and late maturing Genotype 15, are shown in bold italics (unless both appear in a given ranking, in which case only the leader is highlighted).

Environment	AMMI1 Rank					AMMI3 Rank					AMMIF Rank				
	1	2	3	4	5	1	2	3	4	5	1	2	3	4	5
25	5	4	7	9	6	5	4	7	9	17	5	1	4	9	17
8	5	4	7	9	6	5	4	9	15	12	5	4	18	12	15
21	5	4	7	9	6	5	7	4	10	9	9	16	5	7	4
18	5	4	9	7	6	5	7	4	2	8	7	4	5	2	17
12	5	4	9	7	6	5	4	9	7	6	18	7	6	5	12
15	5	4	9	7	6	5	4	9	7	6	5	9	4	1	7
7	5	4	9	7	6	6	4	9	16	5	6	9	16	4	5
22	5	4	9	7	6	5	4	7	9	6	7	1	10	12	4
6	5	4	9	7	6	4	10	7	5	9	5	9	11	7	16
16	5	4	9	7	6	5	7	4	9	6	1	4	18	7	2
9	5	9	4	7	6	5	7	9	4	2	4	5	15	2	6
19	5	9	4	7	6	10	4	9	6	7	2	4	16	18	6
24	5	9	4	7	6	9	4	5	7	6	4	7	2	1	6
17	9	5	4	15	6	6	13	16	9	12	4	11	13	16	6
20	9	5	15	4	14	2	7	14	9	5	8	7	6	9	10
3	9	15	14	2	5	6	9	14	11	4	18	16	6	10	17
10	15	9	14	2	12	2	11	14	10	7	14	10	11	4	15
14	15	9	14	2	11	15	5	9	17	12	15	12	9	18	14
4	15	9	14	2	11	11	14	6	9	2	14	10	9	17	2
5	15	2	14	11	9	15	8	2	17	7	2	15	4	17	5
23	15	2	14	11	9	2	11	14	15	9	16	17	11	2	9
2	15	11	2	14	13	15	2	13	17	11	15	9	8	11	14
1	15	11	2	14	13	15	17	8	2	13	17	8	9	13	15
11	15	11	2	13	14	15	11	13	14	2	14	3	15	12	6
13	15	11	2	13	14	15	13	11	3	14	15	17	1	3	18

roster can be further reduced to a quite manageable 2 winners, Genotypes 5 and 15, with very little loss of yield. This simplified version of AMMI1 is denoted AMMI1S. The 25 environments are listed in order by their IPC1 scores, rather than their order in Table 6 in Crossa et al. (1991), to make this table more structured and informative.

The AMMI1S mega-environments have an evident agricultural interpretation in terms of a gradient from short growing seasons at the top of Table 2 to long growing seasons at the bottom. Correspondingly, Genotype 5 that wins at the top of Table 2 is an early maturing genotype, whereas Genotype 15 that wins at the bottom is a late maturing genotype. This ecological gradient makes extension of this mega-environment scheme to other locations relatively straightforward. The 2 mega-environments delineated by AMMI1S are almost identical to the 2 mega-environments of Crossa et al. (1991). The only exception is that environment 3, which is right on the boundary between the mega-environments in Table 2, is placed here in the mega-environment at the bottom of the ranking table with long growing seasons because the late

maturing Genotype 15 outperforms the early maturing Genotype 5 in this environment.

Crossa et al. (1991) provides a striking illustration of the importance of mega-environments for this international bread wheat trial, focusing on Genotypes 5 and 15, which they name Kauz#2 and Hahn*2/Prl. They grouped genotypes into the top, middle, and bottom third of the entries. Their Fig. 4 shows that across all locations, Genotypes 5 and 15 had mixed results, appearing frequently in both the top and bottom groups. But after subdivision into 2 mega-environments, Genotype 5 is a decisive winner in the short-season mega-environment and decisive loser in the long-season mega-environment, whereas Genotype 15 is the reverse. And their Fig. 2 illustrates that AMMI1 greatly clarifies the rankings for 3 additional genotypes.

Mega-environments provide simplification by AMMI finding general patterns that fit most of the data. Nevertheless, as Crossa et al. (1991, p. 36) warns, “risk of oversimplification should be recognized” since some genotypes and environments may constitute real exceptions to general patterns. For this reason, Table 2 includes AMMIF

Table 3. The relationship between mega-environments and yield. Results are shown for a sequence of increasingly complex models from AMMI0 through AMMI7, plus AMMIF which is the most complex model equaling the raw data and AMMI1S which is intermediate between AMMI0 and AMMI1. The number of mega-environments (M) increases from 1 to 12 with increasingly complex Additive Main effects and Multiplicative Interaction (AMMI) models. The yields (t ha^{-1}) are averages across the 25 environments for the top ranked genotype selected in each environment by the designated AMMI model, but with these 25 yields estimated by the most accurate AMMI3 model, rather than either the designated AMMI model or AMMIF. Yield reaches a peak for AMMI3, whereas more parsimonious models underfit signal and more complex models overfit noise in this Ockham's hill. For comparison, the grand mean is $4.65378 \text{ t ha}^{-1}$.

Model	M	Yield (t ha^{-1})
AMMI0	1	5.01200
AMMI1S	2	5.27787
AMMI1	3	5.30761
AMMI2	6	5.37744
AMMI3	8	5.39302
AMMI4	6	5.28474
AMMI5	10	5.29007
AMMI6	10	5.28843
AMMI7	10	5.27992
AMMIF	12	5.17735

so that other AMMI models can be compared with the raw data to detect exceptionally large discrepancies. For AMMI1 analysis of this wheat trial, Crossa et al. (1991, p. 36) judges that “even distribution of AMMI adjustments through the ... data matrix, without large changes centered on a minority of genotypes or locations, allayed fear of such oversimplification.” The residuals approximate a normal distribution without glaring exceptions. Ideally, the detection of outliers or exceptions relies not only on statistical considerations, but also on extensive knowledge of peculiar disease pressures, soil conditions, or other factors affecting particular environments or genotypes.

Significantly, Crossa et al. (1991, p. 36) judges their 2 mega-environments (which are identical to the AMMI1S mega-environments, apart from a single environment right on the border) to be “reasonably congruent” with the CIMMYT wheat breeding program's well established classification of major agroclimatic areas of wheat production based on extensive information including “climatic conditions, foliar disease complexes, and soil type.” For this challenging international yield trial with diverse genotypes and environments, it is quite impressive and encouraging that AMMI results based on only yield data for only 1 yr can so closely match expertise based on many kinds of data collected over many years.

The combination of accurate yield estimates and ecologically ordered environments from AMMI analysis renders a yield trial much more comprehensible after AMMI analysis than before. This can be seen by contrasting the right and left sides of Table 2. First, consider the yield

trial before AMMI analysis by inspecting the raw data AMMIF rankings at the right, except imagine that the environments are mixed up in a random order (such as the actual original order given at the extreme left in Table 2, which is random from an ecological perspective). Then, compare the AMMI1 rankings at the left resulting from gaining accuracy and ordering environments.

Fourth Step

Selection or recommendation of the best genotypes involves 2 principal considerations. First, selection must inevitably be done in the context of a given mega-environment scheme, whether that scheme be a single mega-environment exploiting only broad adaptations, or else multiple mega-environments exploiting both broad and narrow adaptations. Second, selection should ideally be done with yield estimates as accurate as possible, using both the treatment and experimental designs.

Table 3 shows the relationship between mega-environments and yield. Results are shown for 10 AMMI models, from the most parsimonious AMMI0 to the most complex AMMIF, but not for AMMI8 through AMMI16 because they exhibit little change. Increasingly complex AMMI models implicate a generally increasing number of mega-environments, from 1 to 12. For each model, this table also shows the average yield across the 25 environments for the top ranked genotype in each environment, with these rankings based on yield estimates by the designated AMMI model (as shown in Table 2 for 3 AMMI models of particular interest). However, having used the designated AMMI model to identify the winning genotype in each of the 25 environments, the most accurate member of this model family, which is AMMI3 for this wheat dataset, has been used to estimate the yields of those winners. For instance, AMMI1 winners with their yields estimated by AMMI3 achieve an average yield of $5.30761 \text{ t ha}^{-1}$.

Figure 1 shows Ockham's hill for yield as a function of the number of mega-environments, using the yield values from Table 3 (and yields associated with AMMI models having the same number of mega-environments have been averaged). The peak yield of 5.39302 for AMMI3 is at 8 mega-environments, whereas fewer mega-environments underfit signal and more mega-environments overfit noise. Because the peak of Ockham's hill is fairly broad, a large reduction in the number of mega-environments from 8 to only 2 or 3 imposes a comparatively small reduction in yield.

Regarding Table 3, it may seem more conventional or straightforward to estimate the yields by the raw data AMMIF, instead of AMMI3. But that would be a disaster. Using AMMIF instead of AMMI3 for estimating winners' yields amounts to adding back most of the distracting noise GE_N with its SS of 178.704, which alarmingly is over 3 times the magnitude of the SS of 54.337 associated

with broad adaptations. Estimating yields by the less accurate AMMIF model makes Ockham's hill disappear. The results become deceptive and illusory, depicting a steady yield increase (apart from some small fluctuations) for more complex models, culminating in AMMIF reaching 5.66800, which is inflated by noise. Similar results are shown in Fig. 5 in Gauch and Zobel (1997). Also see Fig. 10.1 and 10.5 in Gauch (2012). A second defective approach would be to use a given AMMI model for both identifying winners and estimating yields. But that would also be a disaster, again failing to reveal Ockham's hill. Yields for AMMI1S through AMMI2 are underestimated and yields for AMMI4 through AMMIF are overestimated, again culminating in 5.66800 for AMMIF.

The results in Table 3 illuminate the relative importance of the three opportunities that plant breeders have for increasing yield: (i) broad adaptations, (ii) narrow adaptations that can be exploited with merely 2 mega-environments, and (iii) additional narrow adaptations with more mega-environments. As a baseline, complete ignorance of both broad and narrow adaptations is represented by the grand mean, which equals the expected average yield were a genotype picked at random in each of the 25 environments. First, exploiting broad adaptations without any subdivision into mega-environments means AMMI0, with Genotype 5 being the winner, giving a yield increment of $5.01200 - 4.65378 = 0.35822$. Second, exploiting narrow adaptations with merely 2 mega-environments (featuring Genotypes 5 and 15) means AMMI1S, giving an additional yield increment of $5.27787 - 5.01200 = 0.26587$. Third, exploiting narrow adaptations with more mega-environments to reach the highest possible average yield means AMMI3, giving an additional yield increment of $5.39302 - 5.27787 = 0.11515$, although it must be admitted that this model's 8 mega-environments are rather impractical. Hence, the total yield increment above the baseline of complete ignorance is 0.73924. Finally, converting these 3 increments to percentages, the opportunities to increase yield are 48.4% from broad adaptations, 36.0% from narrow adaptations with merely 2 mega-environments, and an additional 15.6% from narrow adaptations with more mega-environments. Fortunately, most of the opportunity from narrow adaptations (69.8%) can be exploited with merely 2 mega-environments, and this opportunity is substantial relative to that from broad adaptations.

The values in Table 3 quantify the relative benefits from the yield-trial experiment and the statistical analysis. Again the grand mean represents the baseline of complete ignorance of both broad and narrow adaptations. AMMIF represents the information from the yield-trial experiment, providing a yield increment of 0.52357, although this model's 12 mega-environments are wholly impractical. AMMI1S represents the additional information from the statistical analysis, providing a further yield increment

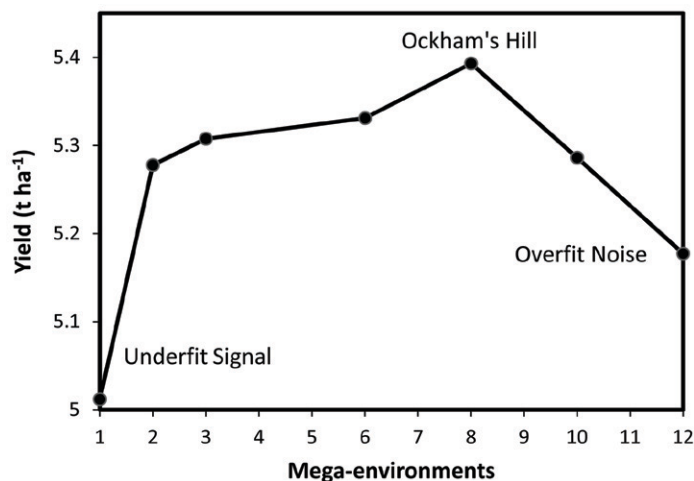


Figure 1. Yield (t ha^{-1}) as a function of the number of mega-environments. The response is an Ockham's hill with the peak yield of 5.39302 for AMMI3 at 8 mega-environments, whereas fewer mega-environments underfit signal and more mega-environments overfit noise. Moving from the left to the right, this spectrum from simple to complex mega-environment schemes corresponds to the spectrum from parsimonious to complex AMMI models. At the extreme left, 1 mega-environment equates to AMMI0 that discards and underfits all of the narrow adaptations in the interaction signal GE_S . At the extreme right, 12 mega-environments equate to the raw data AMMIF that captures and overfits all of the interaction noise GE_N . In between at the peak of Ockham's hill with 8 mega-environments, AMMI3 captures most of GE_S and discards most of GE_N .

of 0.10052, and this model's 2 mega-environments are quite practical. Converting these increments to percentages, 83.9% of the yield gain comes from the experiment and 16.1% from the analysis. Because the cost of the statistical analysis is miniscule compared with the cost of the yield trial, this analysis provides an extremely cost-effective opportunity for adding value to this agricultural research. Furthermore, in the context of plant breeding with recurrent selection involving multi-environment yield trials, a bonus of 15 or 20% from AMMI analysis would *double* yield gains after 5 or 4 cycles of selection.

CONCLUSION

For multi-environment yield trials used in plant breeding, G represents broad adaptations and GE_S represents narrow adaptations, whereas E is irrelevant and GE_N is deleterious. Generally, only a portion GE_p of GE_S can be exploited to increase yields because only predictable environmental factors are useful (associated with locations and management practices but not years), and because practical constraints limit the number of mega-environments (often to only 2 or 3).

Given those agricultural considerations, AMMI provides an ideal statistical analysis. The initial AOV partitions the variation in the yield trial into G, E, and GE, while the subsequent PCA applied to GE selectively

captures GE_S in early components and selectively relegates GE_N to a discarded residual. Furthermore, in the most common case that several IPC (typically 2 to 5) contribute to predictive accuracy but the consequent mega-environments are too numerous to be agriculturally feasible, a lower-order AMMI model (often AMMI1, perhaps further simplified by merging small into large mega-environments) can focus on GE_p , a practical portion of GE_S . AMMI analysis of multi-environment yield trials serves two main purposes: (i) understanding complex genotype $\times$ environment interactions, including delineating mega-environments and selecting genotypes to exploit narrow adaptations, and (ii) gaining accuracy to improve recommendations, repeatability, selections, and genetic gains.

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